

E-Commerce Customer Segmentation and Cohort Retention

UCI Online Retail II, December 2009 to 9 December 2011 · Novaldi Ramadhan Waluyo

Context

The project follows a case brief: a UK online retailer has seen customer acquisition cost rise 25% in 12 months while marketing still relies on untargeted mass promotions. The questions are how many new customers come back after their first order, and which customer groups are worth a retention or win-back effort.

Revenue (identified customers)	Cleaned transactions	Customers	Return in month 1 (avg)
£17.74M	805,549	5,878	21%

Key findings

Most customers do not come back in the first month

Across the 24 cohorts with a month-1 value, on average only 21% of new customers bought again in the month after their first order. Retention stays near 20% for the next few months and is about 18% at month 6 and 18% at month 12. The largest loss happens right after the first order.

A small group of customers brings in most of the revenue

Champions and Loyal Customers are 43% of customers but 85% of revenue. Champions alone (1,324 customers) account for 69% of revenue.

At Risk customers are the clearest win-back target

683 customers who used to buy often (5.3 orders on average) have not ordered for about a year (364 days on average). They spent £1.71M in total.

Revenue is seasonal

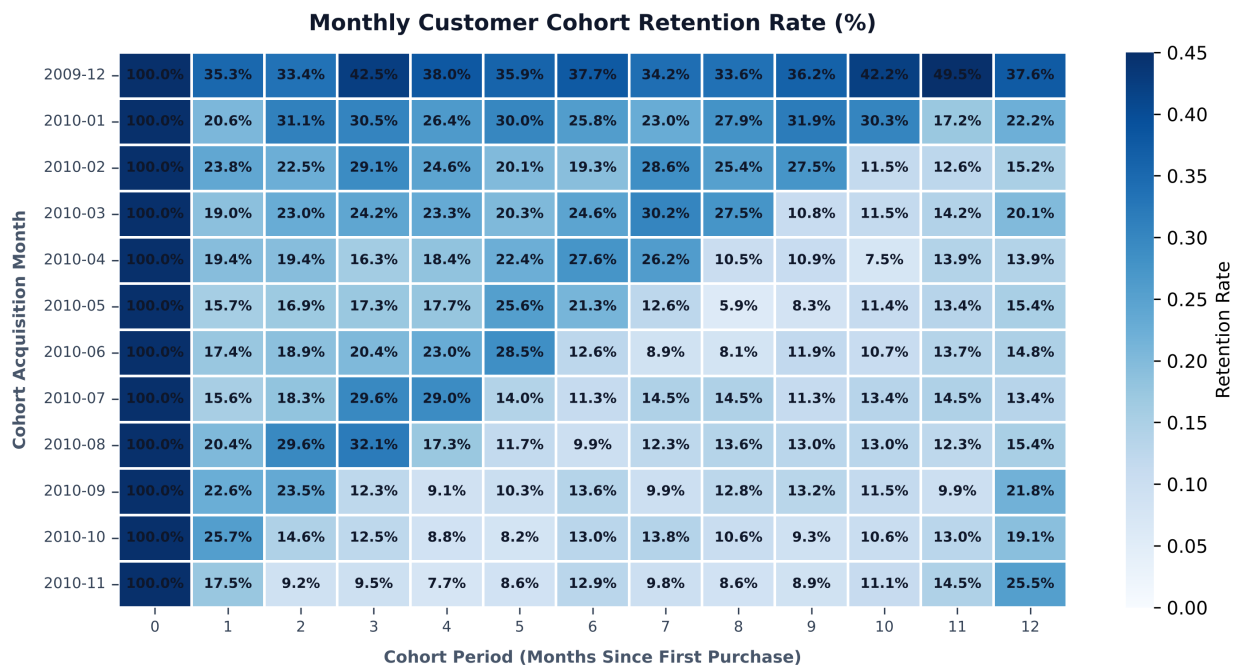
Monthly revenue rises from September and peaks in November in both 2010 and 2011, at roughly twice the level of early 2010. December 2011 looks low only because the data ends on 9 December.

Data and cleaning

1,067,371 raw rows were reduced to 805,549 by removing rows without a customer ID, cancelled invoices (invoice numbers starting with C) and rows with zero or negative quantity or price. Line revenue is quantity × unit price.

1. Cohort retention

Each row is the month of a customer's first order; each column is the number of months since then. Cells show the share of the cohort that ordered again in that month.

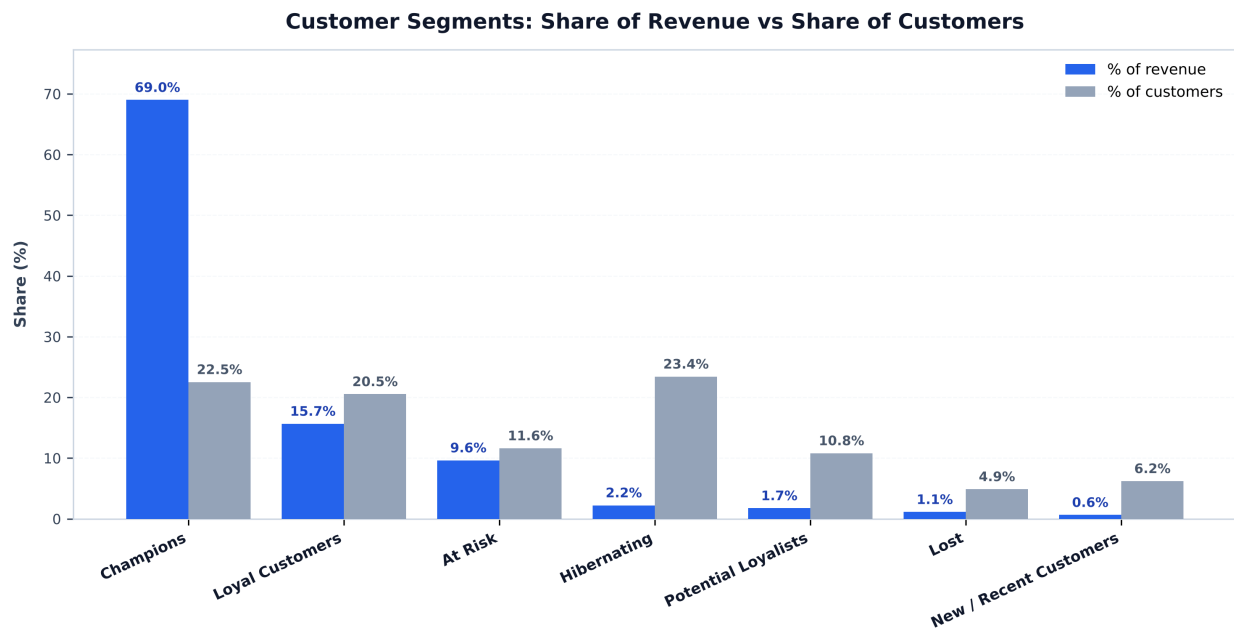


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The December 2009 cohort retains much better than the rest (35% in month 1) because the data starts that month, so it includes customers who were already buying before.

2. RFM segmentation

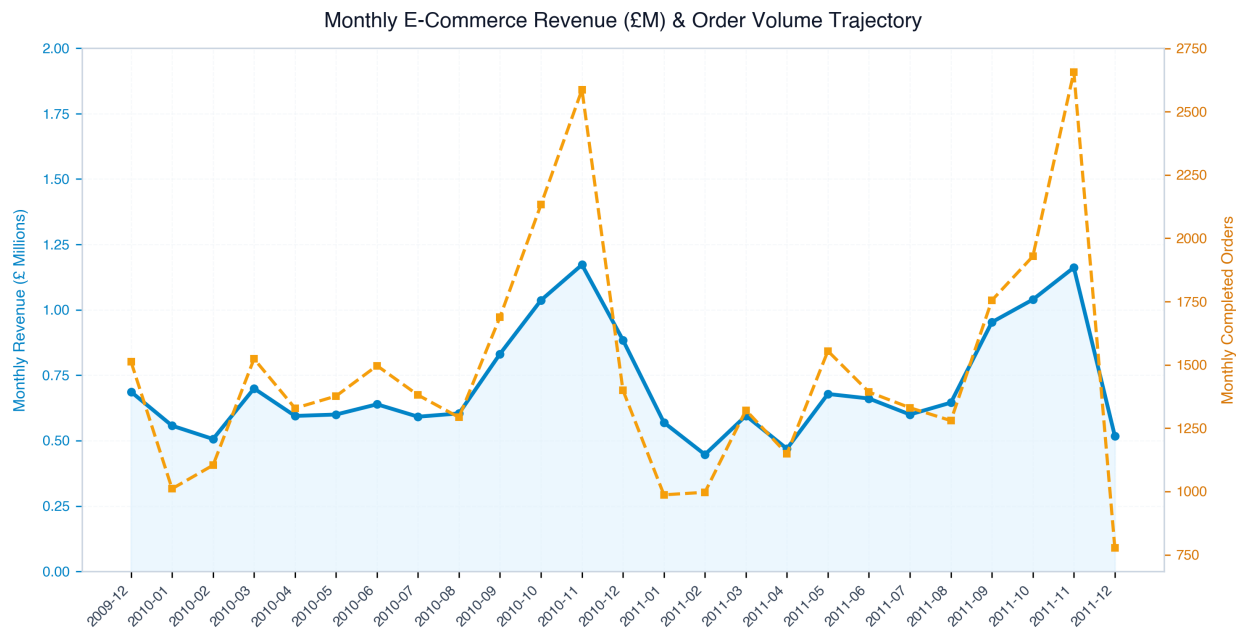
Customers were scored 1 to 5 on recency, frequency and monetary value using quintiles, with 10 December 2011 as the reference date, and then grouped into segments.



Segment	Customers	% customers	Revenue	% revenue	Days since last order	Avg orders
Champions	1,324	22.5%	£12.24M	69.0%	20	16.9
Loyal Customers	1,207	20.5%	£2.78M	15.7%	74	5.8
At Risk	683	11.6%	£1.71M	9.6%	364	5.3
Hibernating	1,377	23.4%	£394,013	2.2%	466	1.2
Potential Loyalists	635	10.8%	£309,190	1.7%	84	1.9
Lost	287	4.9%	£198,631	1.1%	396	2.3
New / Recent Customers	365	6.2%	£112,874	0.6%	29	1.4

Champions and Loyal Customers are 43% of customers but 85% of revenue. Champions alone (1,324 customers) account for 69% of revenue. 683 customers who used to buy often (5.3 orders on average) have not ordered for about a year (364 days on average). They spent £1.71M in total.

3. Seasonality



Monthly revenue rises from September and peaks in November in both 2010 and 2011, at roughly twice the level of early 2010. December 2011 looks low only because the data ends on 9 December.

Recommendations

These are directional, since the data does not include cost or campaign results.

- 1. Focus on the second purchase.** The biggest drop is in the first month, so a follow-up offer or product recommendation a few weeks after the first order targets exactly that gap.
- 2. Run a win-back campaign for At Risk customers.** They have a proven buying history and £1.71M of past spend. Even a small reactivation rate is worth more than a campaign aimed at Hibernating customers (1,377 customers, 2.2% of revenue).
- 3. Keep Champions without heavy discounts.** They already buy often, so early access or service perks are a cheaper way to keep them than price cuts.
- 4. Plan stock and campaigns around September to November.** Demand is highest in these months in both years.

Limitations

- The dataset has no marketing cost, acquisition channel or margin data, so recommendations cannot be costed. The 25% CAC increase comes from the project brief, not from the data.
- Rows without a customer ID (about a quarter of the raw data) are excluded; revenue covers identified customers only.
- The December 2009 cohort retains much better than the rest (35% in month 1) because the data starts that month, so it includes customers who were already buying before.
- The data ends on 9 December 2011, so recent cohorts have only a few months of history.
- RFM segments depend on the chosen score thresholds; different cut-offs would move some customers between segments.